MSc thesis defense

Data-Driven Cognitive Phenotyping in Acquired Brain Injury

A Clustering Analysis with Imputation Robustness Assessment

Zoltan Kunos

MSc in Fundamental Principles of Data Science
Universitat de Barcelona
Supervisor: Dr. Alejandro García-Rudolph

Research question

Do routine cognitive tests reveal distinct patient profiles?

Repeatable score groups exist under specified analytical choices.

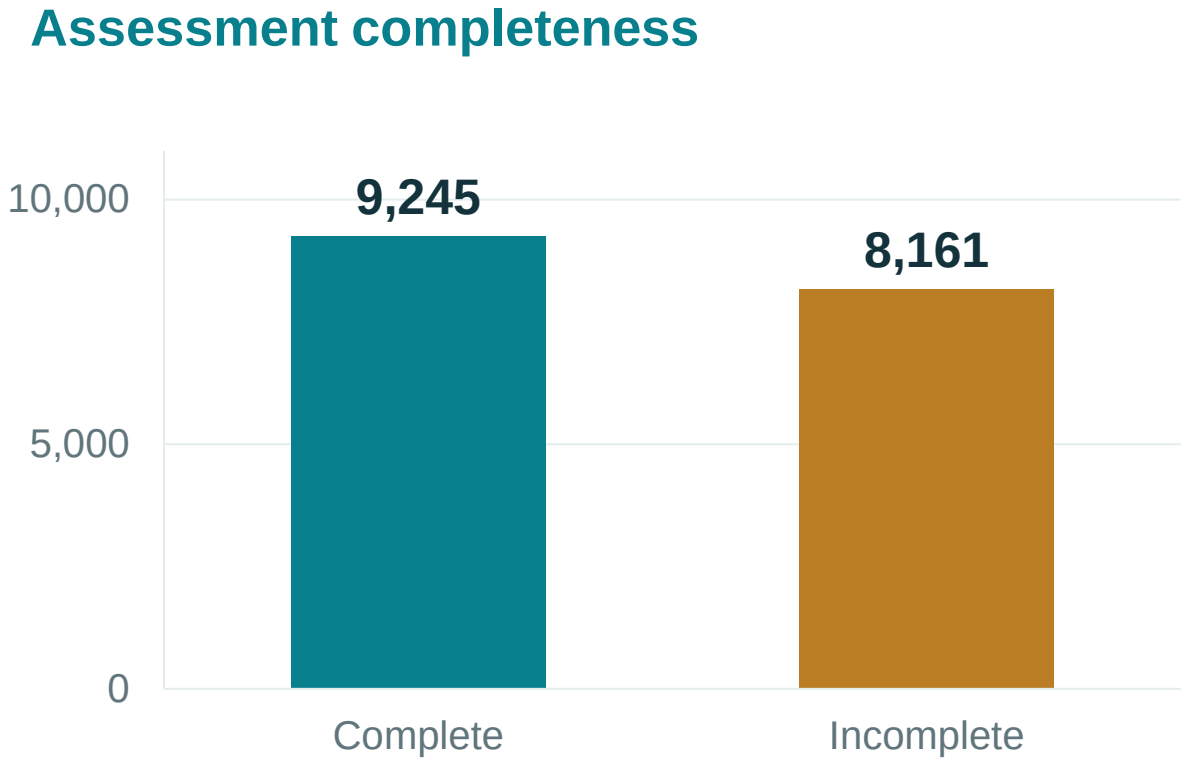
The evidence for distinct clinical phenotypes remains unresolved.

A large cohort with incomplete assessments

17,406

eligible assessments

7,285 patients
14 tests across 6 domains



53.1% of assessments are complete. Repeated assessments are not independent patients.

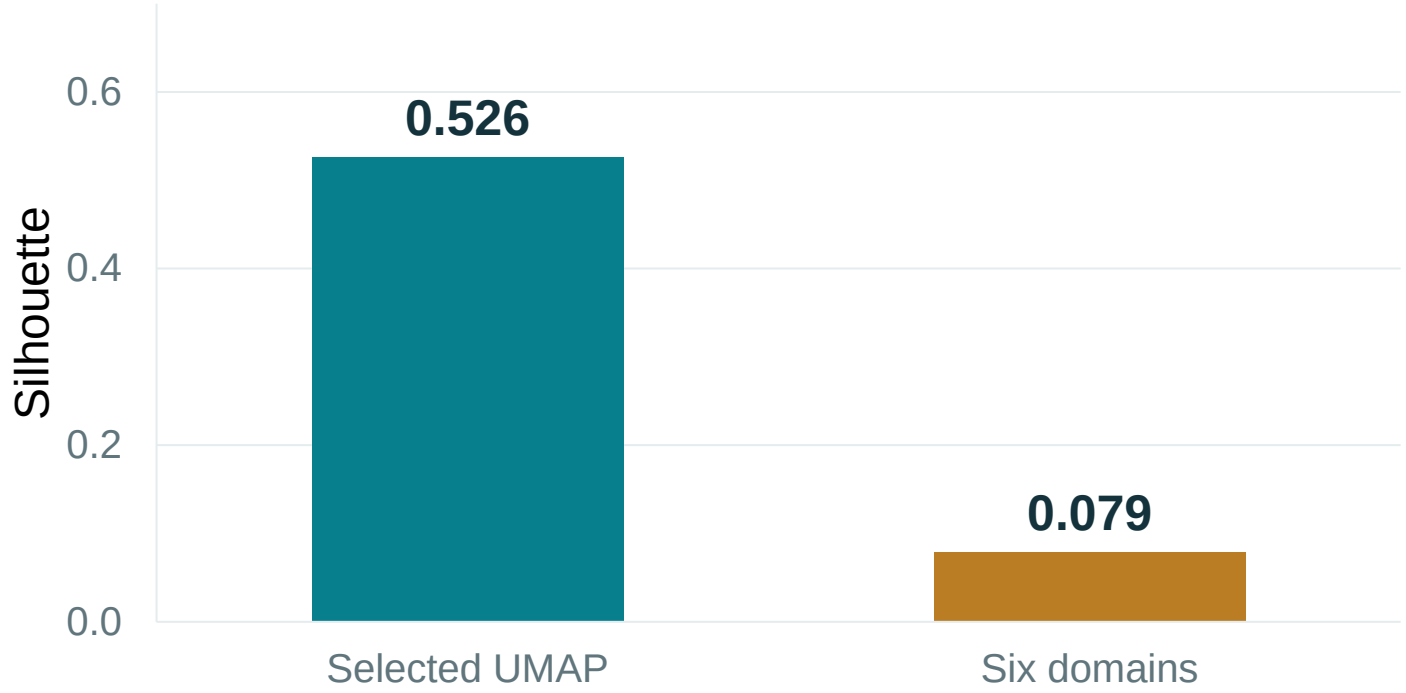
Three analyses answer different questions

Analysis	Unit and design	Question
Historical discovery	17,406 assessments UMAP + HDBSCAN	What group structure appears?
Fixed-reference H2	Same assessments Frozen scaler, embedding and k-means rule	Do imputations change assignments?
Patient extension	7,285 index profiles Independent fits and held-out assignments	Do groups repeat after refitting?

ARI (adjusted Rand index) measures partition agreement. These designs cannot be ranked by ARI alone.

Historical groups mainly differ in score level

The same MICE labels in two spaces



3 groups

6,725
6,328
4,353

assessments per group

Cohort-relative score bands. The evidence does not establish clinically distinct subtypes.

The original hypotheses have qualified support

Hypothesis	Key evidence	Interpretation
H1 Cluster structure	Core Jaccard 0.509–0.630 Cluster count varies from 3 to 5	Partial support
H2 Imputation agreement	Median ARI 0.739 Mean ARI 0.710	Supported within the fixed rule
H3 Clinical-unit association	Cramér’s V 0.162 Added prediction remains uncertain	Association supported
H4 Domain aggregation	Mean VIF 2.77 to 1.88 Silhouette 0.473 to 0.481	Better conditioning

Historical resampling uses assessments, so repeated patients limit uncertainty estimates.

Patient-level discovery and validation

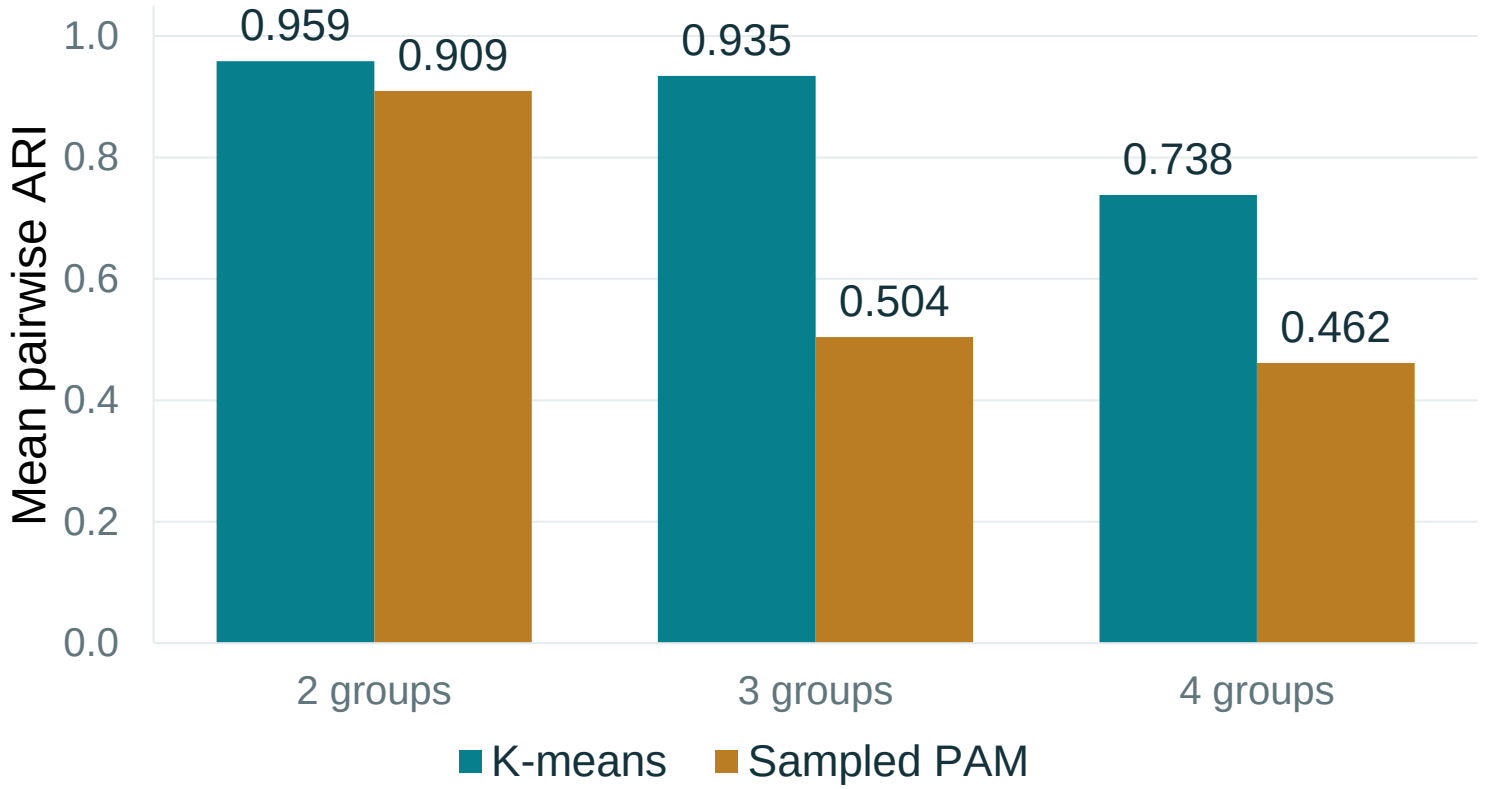
5,828
discovery patients

1,457
validation patients

1. Select one index profile per patient
2. Fit preprocessing, imputation and domain scaling on discovery data
3. Fit each clustering model and assign the validation patients
4. Repeat independent fits and compare the same validation patients

No patient crosses the split. Feature selection was inherited from the full historical cohort.

K-means groups repeat across imputation draws



Additional k = 3 checks

0.968

50 patient subsamples

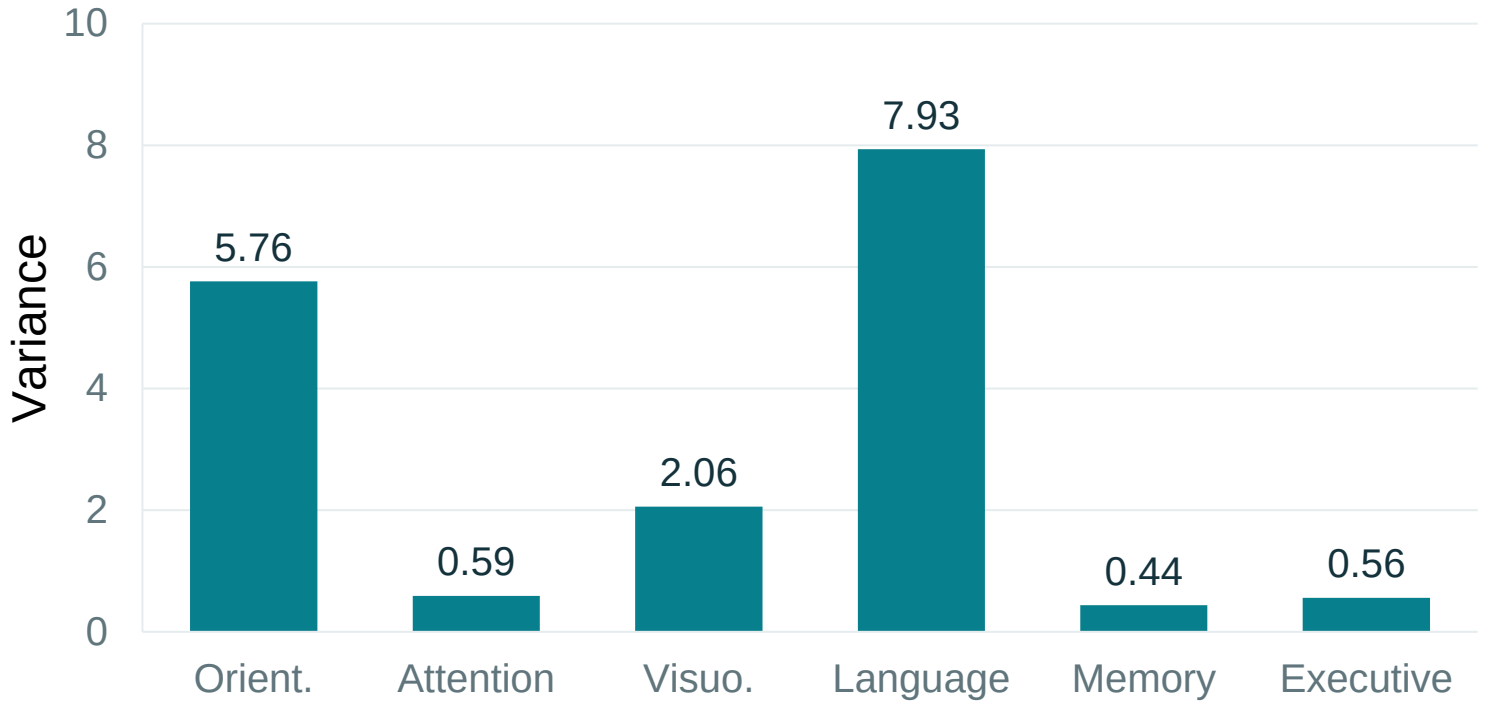
1.000

50 initialization runs

20 stochastic imputations. Agreement is conditional on the imputation model and representation.

Scaling changes who belongs to each group

Discovery variance after robust scaling



0.243

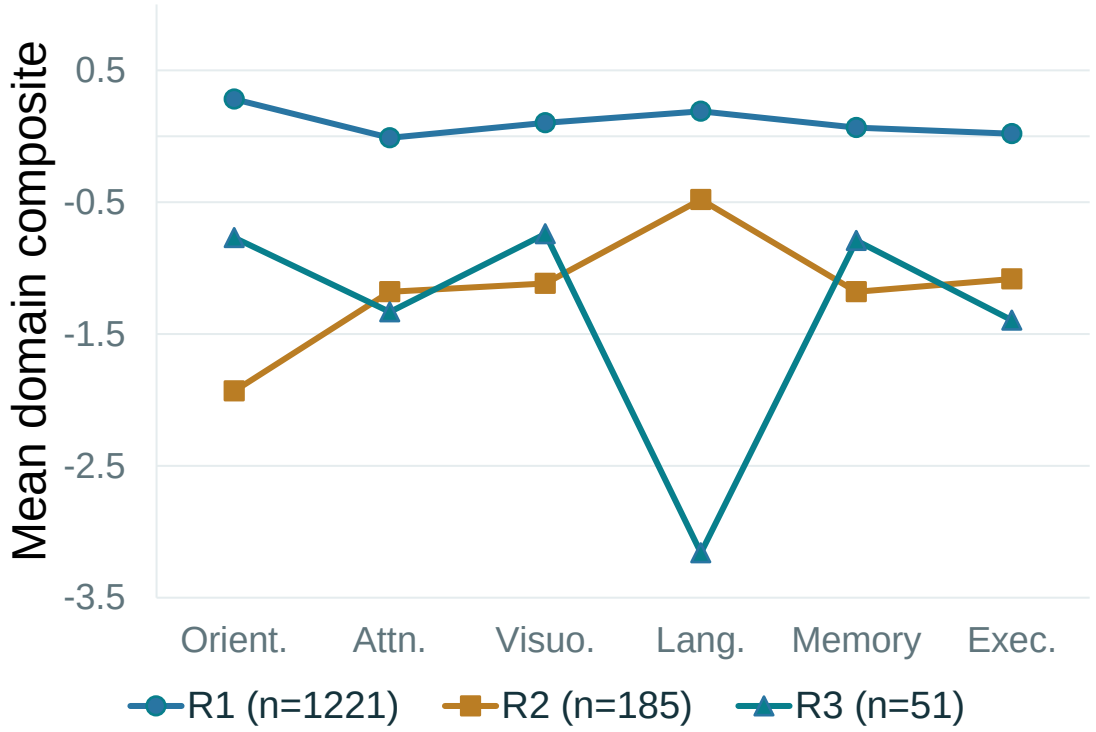
ARI between scaling choices

Same patients
Fixed 3-group model

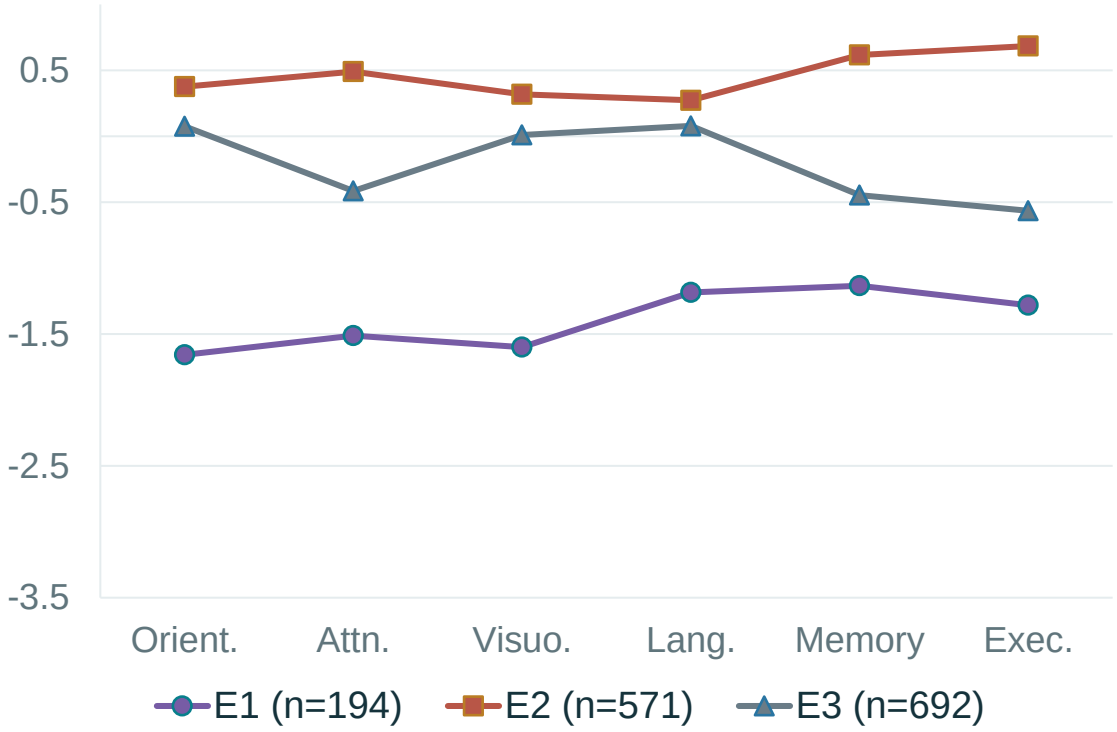
Equal-domain scaling sets every domain variance to 1. Neither weighting has clinical validation.

The alternative scalings produce different profiles

Robust scaling

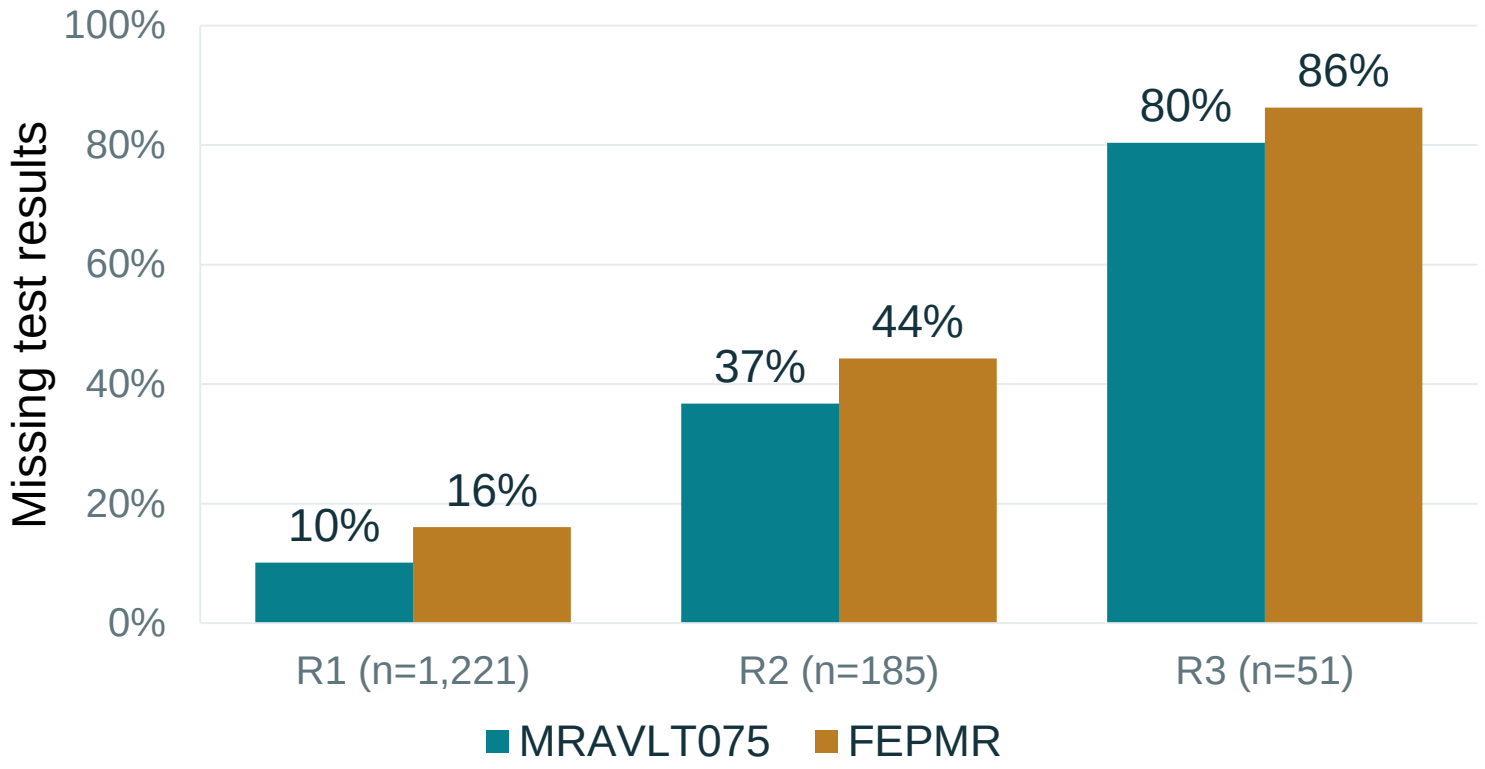


Equal-domain scaling



The vertical scale is shared. Group identifiers and colours are local to each model.

Test availability remains tied to group membership



Imputer sensitivity

0.538

KNN versus reference ARI

Mean imputation: 0.921

Associations are descriptive. They do not identify the cause of the group structure.

Model choice changes the recovered cluster count

Selected model	Groups	Validation silhouette
K-means	2	0.649
Sampled PAM	2	0.540
Tied Gaussian mixture	5	0.535
Full Gaussian mixture	8	0.118
UMAP + HDBSCAN	2	0.400

All rows use robust-domain silhouette and 100% validation coverage. Selection criteria differ.

Alternative models leave subgroup counts unresolved

Analysis	Observed result	Interpretation
14 incomplete tests Observed-data mixtures	Best restricted family: t, df=4 8 components at the search limit	Component count remains unresolved
2,013 longitudinal patients Linear trajectory mixtures	Every multiclass fit contains a class below 5%	One class meets the size rule Recovery need not be uniform

Distributional fit and membership repeatability do not establish clinical utility.

CogDash makes the findings explorable



Research exploration

Saved-model distances

Clinical utility remains untested

Limits that affect the interpretation

External validity	Single-centre cohort and no independent functional outcome
Representation	Inherited feature selection and exploratory domain weighting
Missingness	Agreement under an imputer does not resolve MNAR
Model scope	Restricted mixture families and linear trajectory assumptions

Contribution and next validation step

Contribution

A reproducible analysis of score groups and the choices that change them

Next empirical test

External cohort and functional outcomes
Continuous scores versus added value from group membership

Repeatable score groups

Clinical phenotypes remain unconfirmed

Questions and discussion

Hypothesis criteria and outcomes

Hypothesis	Original criterion	Outcome
H1	At least 2 groups Silhouette ≥0.40, noise ≤30%	10/10 meet embedding criteria Core stability is moderate
H2	Median pairwise ARI >0.50 Fixed reference	Median 0.739 Supported within this design
H3	Association with clinical unit Cramér’s V >0.10	V = 0.162 Added prediction unconfirmed
H4	Similar separation Lower VIF and condition number	VIF 2.77 to 1.88 Condition number 61.7 to 11.8

Fixed-reference agreement and independent refitting

Feature	Historical H2	Patient extension
Observations	17,406 assessments	7,285 index patients
Imputation comparison	10 methods, 45 pairs	20 stochastic Bayesian draws
Transformation and fit	Frozen MICE-based reference	Independent discovery fits
Comparison set	Same assessment rows	Same 1,457 validation patients
Three-group mean ARI	0.710	0.935

Historical assessment subsampling: mean ARI 0.273. This tests a different source of variability.

Independent imputation agreement in full

Method	Groups	Mean ARI	5th–95th percentiles
K-means	2	0.959	0.944–0.974
K-means	3	0.935	0.917–0.951
K-means	4	0.738	0.627–0.823
Sampled PAM	2	0.909	0.851–0.959
Sampled PAM	3	0.504	0.281–0.878
Sampled PAM	4	0.462	0.308–0.662

Percentiles describe dependent comparisons. They are not confidence intervals.

Partition changes and geometry changes are separable

Fixed validation labels	Scored in robust space	Scored in equal-domain space
Robust k = 3	0.608	0.440
Equal-domain k = 3	0.172	0.283

ARI between the two partitions: 0.243

A silhouette is conditional on geometry. The same labels can receive different scores.

Incomplete-data mixture comparison

Family	k	BIC	Validation log likelihood	Domain silhouette
Gaussian	6	73878.4	-6.076	0.188
Contaminated Gaussian	7	74641.8	-6.030	0.154
t, df = 10	8	69815.8	-5.550	0.111
t, df = 4	8	67545.6	-5.341	0.095

28 of 32 fits converged

Likelihoods refer to the same 14-test representation. The upper-bound optimum remains unresolved.

Continuous-reference checks give mixed evidence

Diagnostic	Fitted-reference tail fraction	Meaning within this reference
Maximum silhouette	0.0099	More separation than the fitted copula
PC1 dip statistic	0.9901	No excess projected multimodality

1,000 patients and 100 reference simulations

Descriptive calibration. Missingness masks are independent of simulated values.

The trajectory size rule selects one class

Classes	BIC	Validation log likelihood	Smallest expected class
1	8376.3	-2.598	100.0%
2	7572.2	-2.403	3.9%
3	6806.0	-2.290	2.7%
4	6465.5	-2.212	1.3%

2,013 patients: 1,590 discovery and 423 validation

Every class must contain at least 5% expected discovery membership. Nonlinear change remains open.

Preprocessing sensitivity at fixed three-group k-means

Change from reference	Validation patients	ARI versus reference
Mean imputation	1,457	0.921
KNN imputation	1,457	0.538
At least 7 observed tests	1,339	0.903
No clipping	1,457	0.958
50-iteration imputation limit	1,457	1.000
Independent validation clustering	1,457	0.895

The final row measures partition agreement under a frozen transformation, not predictive accuracy.

Algorithms and missing-data assumptions

Historical imputation comparison

Mean, EM, MICE, KNN, MissForest, PMM, SoftImpute,
NMF, denoising autoencoder and variational autoencoder

Historical discovery settings

UMAP: 8 dimensions, 15 neighbours, min_dist = 0
HDBSCAN: minimum cluster size 1,000 and min_samples = 50

The patient extension uses a separate bounded Bayesian imputer. MAR/MNAR uncertainty remains.

Clinical association and practical value

0.162

Cramér’s V for clinical-unit association

0.0013

Log-loss improvement beyond diagnosis

95% patient-bootstrap interval: −0.0006 to 0.0031

The interval includes zero. No independent functional outcome was available.